Cayleb Alvarez-James

AI Agent Systems Engineer - Claude Code & Coding-Agent Tooling

Middletown, Connecticut, USA | caylebalvarezjames@gmail.com

github.com/cayleb-james2008 | linkedin.com/in/cayleb-alvarez-james-99850531b | cayleb-james2008.github.io/agentic-resume/

Summary

Technical founder who builds with Claude Code every day. I run fleets of AI agents - dynamic swarms that write, check and ship real software under rules I designed, where a job counts as done only when an independent check passes on the real thing. I have put agents to work inside a real company (RHL, a furniture importer) and run a live trading house where every strategy earns its place on measured results. Built all of this while working as a furniture installer. 1,200+ GitHub contributions in the last year; 4,000+ automated tests across five shipped systems.

Experience

Founder - Independent AI Agent Systems | Oct 2025-present

Created Ergon, piloted inside RHL; built and run the DSH Fleet harness and OpenBell, a live automated trading house. Harness changes ship only when a fresh checker measures them beating the current version on a fixed scorecard.

Selected Projects

Ergon - Personal AI Agents for Companies

Node.js, C#, Python, SQLite

Ergon case study

- Self-configuring, autonomous role-learning agent: after install it observes the employee's recurring work, proactively asks targeted questions, builds its own model of the role, and iteratively proposes, builds and vets capabilities - the deployer does not manually monitor, adapt, configure or pre-map the employee's job. One personal AI agent per employee, built around each company's own tools, with human approval gates and undo on anything that matters.
- Piloted inside a real company: the warehouse operations board built for RHL runs on-site in daily use. Signed Windows installer and taskbar app; ticket-scoped remote repair and connectors to existing company apps. 622 automated tests pass.

DSH Fleet Harness - Multi-Agent Orchestration & Verification

Python, TypeScript

github.com/cayleb-james2008/dsh-fleet-genesis

- Ticket-driven harness with no standing bench: each ticket or goal spins up a dynamic swarm shaped to the task - it decomposes the goal, spawns its own sub-agents, does research, authors or reuses skills, verifies the work and dissolves.
- Distributed, recursively self-improving infrastructure: swarms revise the fleet's own workflows, plugin behaviour, Omni Loop and skill library from measured job outcomes. It spans desktop, Windows VM, phone VM and browser execution surfaces, with Slack as its bring-your-own-agent coordination layer for outside agents. Guided by a harness-engineering research base, with human approval gates and a self-improvement loop gated by measured results. In active development.

OpenBell - Live Trading House: Slot Method & Weight-Space Models

Python, Rust

OpenBell case study

- Weight-space programme: instead of training black-box models, I directly construct and author tiny model weights (linear specialists), freeze the prediction in writing before scoring, and evaluate under strict chronological, cost-aware, kill-gated evidence rules - no money test until the training win rate clears the breakeven for its exit shape. Conventional distillation failed on both test instruments; the hand-authored weights passed, and the best configuration was net-positive on historical data.
- All of my trading and finance work under one house: Kalshi prediction markets, a futures desk and a meme-coin desk - the controlled proving ground for the weight-space work, run on one architecture of parallel strategy slots splitting the equity evenly, each slot iterating on measured results and getting wiped and replaced when it fails. Live desks run small real capital with reconciled accounting and pre-registered kill conditions.

Sophos - Coding-Agent Desktop App

React, TypeScript, Tauri

github.com/cayleb-james2008/sophos

- Desktop app that runs coding agents from saved profiles, with sessions and transcripts that survive restarts and clear errors when a provider fails.
- 1,224 interface tests, 82 Node tests and 56/56 Windows checks pass; merged after full hosted CI.

dotz - AI Coding Agent Dashboard

Rust, Tauri, JavaScript, SQLite

github.com/cayleb-james2008/dotz

- Desktop app that runs and supervises AI coding agents, with model choice and safe process-tree cleanup.
- 828+ core tests pass on each of Windows, macOS and Linux; all 6 hosted check jobs pass; merged.

Asmodeus - Trading Research on Past Data

Rust

Asmodeus case study

- Tests trading ideas on real historical data in strict time order, with fees and safeguards against peeking at future data; rejected ideas are logged, not quietly dropped. 1,111 Rust tests pass. Research only - no real money.

Industry AI Suite - Source-Checked Review Tools

Python

github.com/cayleb-james2008/industry-ai-suite | Review history implementation

- Ten local tools that review finance, research, security and operations files; every finding shows its source and a person decides. Drafts halt when sources are missing. 382 Python tests pass.

Skills

AI & agents: Claude Code (daily, primary tool), LLM agents, multi-agent orchestration, evaluation and verification design, human approval gates, adversarial testing of agent output

Languages: Python, TypeScript, JavaScript, Rust, HTML/CSS

Apps & data: React, Tauri, SQLite, web APIs, Git/GitHub, Linux

Additional Experience

Robert H. Lord Company - Installer & AI Integration | Sep 2026-present

David Cooke Plaster Co. - Plaster Work | Summer 2026

Dunkin' | 2024-2026

Tommy's Pizza | Summer 2024

Education

Portland High School | Portland, Connecticut | Graduated 2026

Coursework: Cybersecurity; AP Computer Science